

Occupation Hazards in Dentistry

Lab 10

Dental professionals are at risk for exposure to numerous **biological, chemical, environmental, physical, and psychological workplace hazards.**

We can define "**Occupational hazard**" as a risk to a person usually arising out of employment.

Many occupational health problems still persist in modern dentistry ,These include ;
exposure to infectious diseases, radiation, dental materials, musculoskeletal disorders, dermatitis and respiratory disorders, eye injuries, and psychological problems.

Major occupational hazards are:

1. Biological health hazards
2. Physical hazards
3. Chemical hazards
4. Musculoskeletal disorders and diseases of the peripheral nervous system
5. Hearing loss
6. Radiation exposure
7. Stress
8. Legal hazards
9. Other risks

Biological Health Hazards

These hazards are constituted by infectious agents of human origin which include viruses, bacteria and fungi.

All of the dental team are at risk of exposure to hepatitis **B virus (HBV)**, **HIV infection**, and other types of communicable infections.

Physical Hazards

The dentist and the clinical staff are at risk of physical injuries during many dental procedures. Sources of physical injury can include debris from the oral cavity striking the eyes, cuts from sharp instruments, or puncture wounds from needles or other sharp instruments. Those injuries can result in the transmission of serious infectious disease to the dental worker.

Chemical Hazards

Many of these chemicals are among those whose health effects may not be known and may pose health problems taking years to manifest. Hazardous chemical agents used in clinical dentistry include mercury, powdered natural rubber latex (NRL), disinfectants, and nitrous oxide (N₂O).

Musculoskeletal Disorders and Diseases of the Peripheral Nervous System

At work, the dentist assumes a strained posture (both while standing and sitting close to a patient who remains in a sitting or lying position), which causes an overstress of the spine and limbs. The overstress negatively affects the musculoskeletal system and the peripheral nervous system above all, it affects the peripheral nerves of the upper limbs and neck .

Hearing Loss

The noise of suctions, saliva ejectors, turbines, engines, amalgamators, compressors, etc. may causes impaired hearing. Still some dentist may be at risk especially where faulty or older equipment is used.

Radiation Exposure

The radiation dose received by an individual is generally low and relatively few cells are damaged during an average radiological examination. Thus, the effect of even low levels of exposure to ionizing radiation over periods of time may accumulate and could represent a potential hazard to health.

Dental staff should take steps to protect themselves during exposure by standing behind protective barrier, use of radiation monitoring badges and regular equipment checks and maintenance. Use of safety shields are recommended as they are protective.

Legal Hazards

There are relevant statutes and regulations which apply to the practice of dentistry in every country. The contravention of any of these may warrant that legal actions be brought against a dental practitioner particularly in developed countries where the citizens appear more aware of their rights.

To help assure a safe work environment in dental treatment, the hazard awareness and prevention of legal risks should be made known to all clinical workers of the dental hospital/clinic.

Prevention of Occupation Hazards:

Education, workers must understand any possible negative health implications of their jobs and how to minimize them.

Infection control and proper handling of potentially infected materials.

Barrier techniques include gloves, masks, protective eye wear, high power suction and good ventilation to reduce aerosols and vapor dangers

Prevention of Occupation Hazards:

Hypoallergenic nonlatex gloves are proposed can decline latex allergy

prevent radiation hazard Lead aprons, periodic maintenance of the X ray machine and radiation level sensors

Prevent musculoskeletal disorders: identify symptoms as soon as they become apparent ,Modify working positions to achieve optimal body posture
Achieve optimum access, visibility, comfort, and control at all times

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**THANK
YOU**